LE NGUYEN GIA HUNG

AI Undergraduate | Machine Learning, Computer Vision & Speech
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EDUCATION

FPT University — B.Eng. in Artificial Intelligence Expected Dec 2027
GPA: **8.62/10.0**. Certificate of Merit: semesters 3, 4, 5 and 6.

TECHNICAL SKILLS

Languages & data: Python, SQL/PostgreSQL, Pandas, NumPy.
Machine learning: PyTorch, Transformers, TensorFlow/Keras, Scikit-learn, OpenCV.
Evaluation & tools: SHAP, Integrated Gradients, DiCE, chronological cross-validation; Git, Docker, Linux, Weights & Biases, Streamlit, FastAPI.

SELECTED RESEARCH & PROJECTS

SmokeNet: Air Quality Forecasting Jun – Nov 2025
First & Corresponding Author; published at AJCAI 2025

- Designed a Transformer with seasonal embeddings and reconstruction loss for $PM_{2.5}$ forecasting; built a chronological evaluation pipeline on **4+ years** of OpenWeather data.
- Reported **MAE 0.7470**, R^2 **0.9545** and **35.9% lower MAE than XGBoost** on the 2024 held-out evaluation; used SHAP, Integrated Gradients and DiCE to inspect forecasts. Case study.

MERR-GAT: Multimodal Emotion Recognition Oct 2025 – Present
Second & Corresponding Author; research in progress

- Develop **WavLM + Compact Conformer** audio encoding, BERT/RoBERTa text representations and dual-channel graph-attention fusion.
- Evaluate on IEMOCAP, RAVDESS and MELD using WA, UA and Macro-F1; experiments are ongoing. Methods and code links.

Vietnamese Speech Recognition Sep 2025 – Present
Co-Researcher & Co-Author; research in progress

- Build a **5.4M-parameter** CNN–residual BiLSTM model with CTC, trained on **745 hours** of speech. Reported **WER 33.09%** and **CER 15.28%**; compare with PhoWhisper. Evaluation notes.

PPE Detection — YOLOv8, OpenCV, Streamlit Course project

- Built **9-class** detection, safe/unsafe compliance logic and a webcam interface with violation reporting. Project details.

LEADERSHIP & CLOUD LEARNING

SpeedyLabX, FPT University — Founder & Student Lead Jan 2025 – Present

- Lead a **10-member** undergraduate group with weekly technical workshops, paper reading and shared conference-submission workflows.

First Cloud AI Journey — FCAJ Cloud Intern, Team NeuraX 2026
AWS Vietnam program; cloud learning and project work documented in the program journal.

PUBLICATION

Gia-Hung Nguyen Le et al. "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *AJCAI 2025, Springer LNAI*. First and corresponding author. DOI.
Three additional manuscripts under review; full titles and contributions in the research CV.